

# Zomato Restaurant Data Analysis Report

## Title

Exploratory Data Analysis of the Zomato Restaurant Dataset

## 1. Objective

The objective of this project is to analyze the Zomato restaurant dataset to discover useful insights about restaurants, customer ratings, cuisines, restaurant locations, and pricing. The analysis helps understand customer preferences and factors affecting restaurant ratings.

## 2. Dataset Description

The dataset contains

|             |  |
|-------------|--|
| information |  |
|-------------|--|

about restaurants listed on Zomato, including:

- Restaurant Name
- Address
- Location
- Online Order Availability
- Table Booking Availability
- Ratings
- Votes
- Restaurant Type
- Cuisine
- Approximate Cost for Two
- Popular Dishes

The dataset originally contains **56,252 records** and **13 features**.

## 3. Data Preprocessing

The following preprocessing steps were performed:

- Loaded the dataset using Pandas.
- Checked dataset information and data types.
- Identified missing values.
- Removed duplicate records.
- Replaced missing values with "Unknown".
- Converted restaurant ratings from string format (e.g., "4.1/5") into numeric values for analysis.

## 4. Exploratory Data Analysis

### A. Missing Value Analysis

Several columns contained missing values.

The **Dish Liked** column had the highest number of missing values, followed by **Rate** and **Listed\_in(type)**.

### B. Duplicate Records

The dataset contained **15,703 duplicate records**, which were removed before analysis.

### C. Rating Distribution

A histogram was generated to visualize restaurant ratings.

#### Observation

- Most restaurants have ratings between **3.5 and 4.2**.
- Very few restaurants have ratings below 2.5.

- Highly rated restaurants (above 4.5) are relatively rare.

## D. Top 10 Cuisines

A bar chart was created to identify the most common cuisines.

### Observation

- North Indian cuisine appears most frequently.
- Other popular cuisines include:
  - Chinese
  - South Indian
  - Fast Food
  - Biryani
  - Cafe
  - Bakery

These cuisines dominate restaurant listings.

## E. Price vs Rating

A scatter plot was created between approximate cost for two people and restaurant ratings.

### Observation

- Higher-priced restaurants do not always receive better ratings.
- Many affordable restaurants also achieve excellent ratings.
- Price alone is not a strong predictor of customer satisfaction.

## F. Correlation Heatmap

A correlation heatmap was generated for numeric variables.

### Observation

- Numeric features show relatively weak correlations.
- Restaurant ratings appear to depend on multiple factors rather than a single numeric variable.

## G. Top Restaurant Locations

The top restaurant locations were identified using a bar chart.

### Observation

Popular restaurant hubs include:

- BTM
- Koramangala
- Indiranagar
- Whitefield
- HSR Layout

These locations contain a high concentration of restaurants.

## H. Cuisine Word Cloud

A word cloud was generated from cuisine names.

### Observation

The largest words represent the most frequently served cuisines, such as:

- North Indian
- Chinese
- South Indian
- Fast Food
- Biryani
- Cafe

This provides a quick visual summary of cuisine popularity.

## 5. Key Findings

- The dataset initially contained over **56,000 restaurant records**.
- More than **15,000 duplicate records** were identified and removed.

- Restaurant ratings are generally concentrated between **3.5 and 4.2**.
- North Indian cuisine is the most common cuisine.
- Affordable restaurants can achieve ratings comparable to expensive restaurants.
- Restaurant density is highest in popular commercial areas such as BTM and Koramangala.
- Missing values were most significant in the **Dish Liked** column.

## 6. Business Recommendations

1. Restaurants should maintain consistent food quality to improve customer ratings.
2. Focus on popular cuisines such as North Indian and Chinese to attract more customers.
3. Affordable pricing combined with quality service can be as successful as premium pricing.
4. Restaurant owners should encourage customers to leave ratings and reviews to improve visibility.
5. Expansion efforts should prioritize high-demand locations with strong customer traffic.

## 7. Conclusion

This exploratory data analysis provides valuable insights into restaurant trends within the Zomato dataset. After cleaning the data and removing duplicates, several visualizations highlighted customer preferences, popular cuisines, restaurant distribution, and pricing patterns. The findings indicate that customer satisfaction is influenced by multiple factors beyond price alone. These insights can assist restaurant owners and business analysts in making informed decisions regarding menu offerings, pricing strategies, and location planning.